9

More about Correlation

"Very true," said the Duchess: "flamingoes and mustard both bite. And the moral of that is—'Birds of a feather flock together.'"
"Only mustard isn't a bird," Alice remarked.
"Right, as usual," said the Duchess: "what a clear way you have of putting things!"

—Alice in Wonderland

1. FEATURES OF THE CORRELATION COEFFICIENT

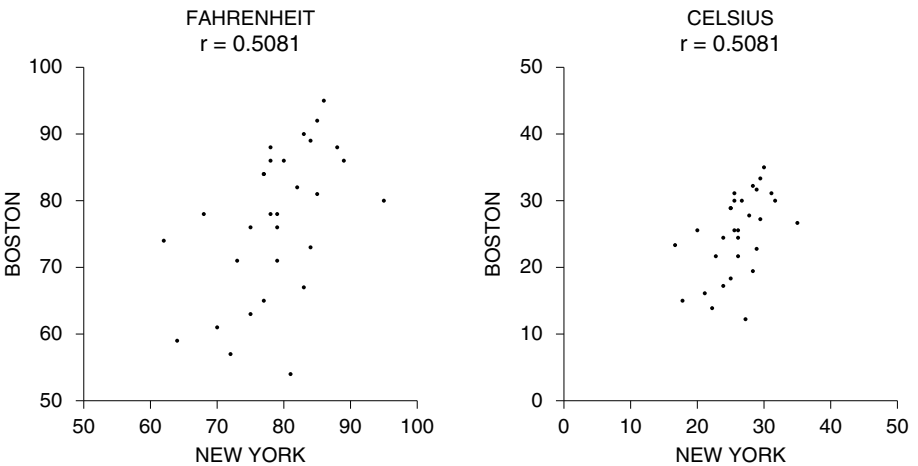
The correlation coefficient is a pure number. Why? Because the first step in computing r is a conversion to standard units. Original units—like inches for height data or degrees for temperature data—cancel out. In a similar way, r is not affected if you multiply all the values of one variable by the same positive number, or if you add the same number to all the values of one variable. (As a statistician might say, r is not affected by *changes of scale*; see pp. 92–93.)

For example, if you multiply each value of x by 3, then the average gets multiplied by 3. All the deviations from average get multiplied by 3 as well, and so does the SD. This common factor cancels in the conversion to standard units. So r stays the same. For another example, suppose you add 7 to each value of x. Then the average of x goes up by 7 too. However, the deviations from average do not change. Neither does r.

Figure 1 (on the next page) shows the correlation between daily maximum temperatures at New York and Boston. There is a dot in the diagram for each day of June 2005. The temperature in New York that day is plotted on the horizontal axis; the Boston temperature, on the vertical. The left hand panel does it in degrees

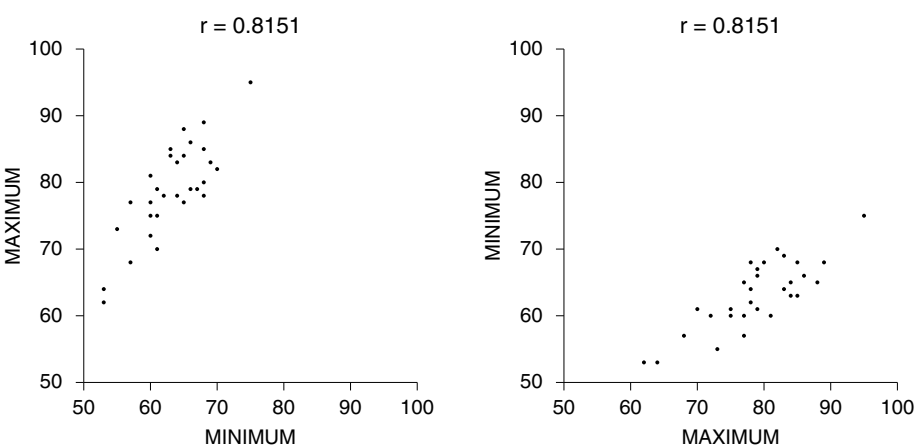
Fahrenheit, and $r = 0.5081$. The right hand panel does it in degrees Celsius, and r stays the same.[1] The conversion from Fahrenheit to Celsius is just a change of scale, which does not affect the correlation.

Figure 1. Daily maximum temperatures. New York and Boston, June 2005. The left hand panel plots the data in degrees Fahrenheit; the right hand panel, in degrees Celsius. This does not change r.



Another feature: The correlation between x and y is the same as the correlation between y and x. For example, the left hand panel in figure 2 is a scatter diagram for temperature data at New York in June 2005. The minimum tempera-

Figure 2. Daily temperatures. New York, June 2005.



ture each day is plotted on the horizontal axis; the maximum, on the vertical. The correlation between the minimum and the maximum temperature is 0.8151. The right hand panel shows exactly the same data. This time, the minimum is plotted on the vertical instead of the horizontal. The pictures look different because the points are reflected around the diagonal. But r stays the same. Switching the order of the variables does not affect r. Why? Remember, r is the average of the products after conversion to standard units. Products do not depend on the order of the factors ($a \times b = b \times a$). It may be surprising that the correlation is only 0.8151, but the weather is full of surprises.

> The correlation coefficient is a pure number, without units. It is not affected by
>
> - interchanging the two variables,
> - adding the same number to all the values of one variable,
> - multiplying all the values of one variable by the same positive number.

Exercise Set A

1. (a) In June 2005, which city was warmer—Boston or New York? Or were they about the same?
   (b) In the left hand panel of figure 2, all the dots are above the 45-degree line. Why?

2. A small data set is shown below; $r \approx 0.76$. If you switch the two columns, does this change r? Explain or calculate.

x	y
1	2
2	3
3	1
4	5
5	6

3. As in exercise 2, but you add 3 to each value of y instead of interchanging the columns.

4. As in exercise 2, but you double each value of x.

5. As in exercise 2, but you interchange the last two values (5 and 6) for y.

6. Suppose the correlation between x and y is 0.73.
   (a) Does the scatter diagram slope up or down?
   (b) If you multiply all the values of y by -1, would the new scatter diagram slope up or down?
   (c) If you multiply all the values of y by -1, what happens to the correlation?

7. Two different investigators are working on a growth study. The first measures the heights of 100 children, in inches. The second prefers the metric system, and changes the results to centimeters (multiplying by the conversion factor 2.54 centimeters per inch). A scatter diagram is plotted, showing for each child its height in inches on the horizontal axis, and height in centimeters on the vertical axis.
   (a) If no mistakes are made in the conversion, what is the correlation?
   (b) What happens to r if mistakes are made in the arithmetic?
   (c) What happens to r if the second investigator goes out and measures the same children again, using metric equipment?

8. In figure 1 on p. 120, the correlation is 0.5. Suppose we plot on the horizontal axis the height of the paternal grandfather (not the father); the height of the son is still plotted on the vertical axis. Would the correlation be more or less than 0.5?

9. Two weathermen compute the correlation between daily maximum temperatures for Washington and Boston. One does it for June; the other does it for the whole year. Who gets the bigger correlation? ("Washington" is the city, not the state.)

10. Six data sets are shown below. In (i), the correlation is 0.8571, and in (ii) the correlation is 0.7857. Find the correlations for the remaining data sets. No arithmetic is necessary.

(i)		(ii)		(iii)		(iv)		(v)		(vi)	
x	y	x	y	x	y	x	y	x	y	x	y
1	2	1	2	2	1	2	2	1	4	0	6
2	3	2	3	3	2	3	3	2	6	1	9
3	1	3	1	1	3	4	1	3	2	2	3
4	4	4	4	4	4	5	4	4	8	3	12
5	6	5	6	6	5	6	6	5	12	4	18
6	5	6	7	7	6	7	5	6	10	5	21
7	7	7	5	5	7	8	7	7	14	6	15

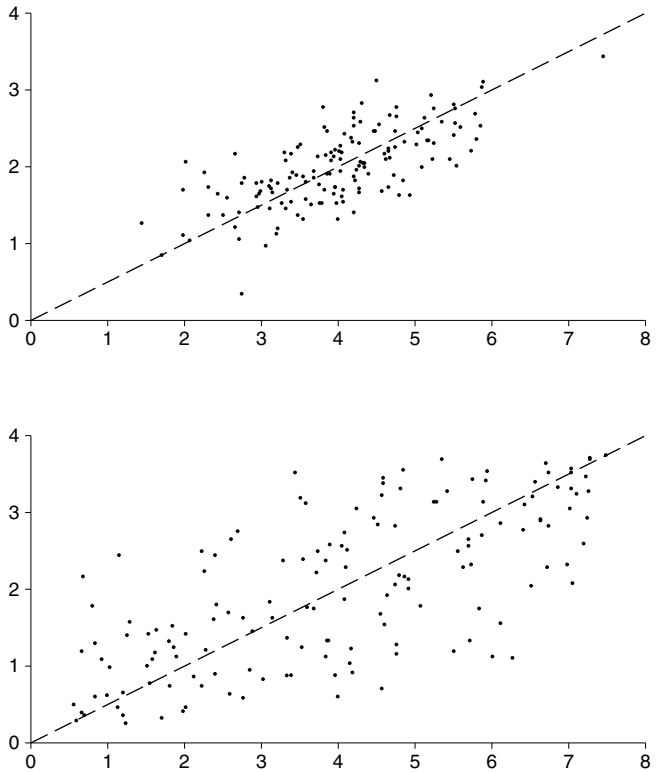
The answers to these exercises are on pp. A57–58.

2. CHANGING SDs

The appearance of a scatter diagram depends on the SDs. For instance, look at figure 3. In both diagrams, r is 0.70. However, the top one looks more tightly clustered around the SD line. That is because its SDs are smaller. The formula for r involves converting the variables to standard units: deviations from average are divided by the SD. So, r measures clustering not in absolute terms but in relative terms—relative to the SDs.

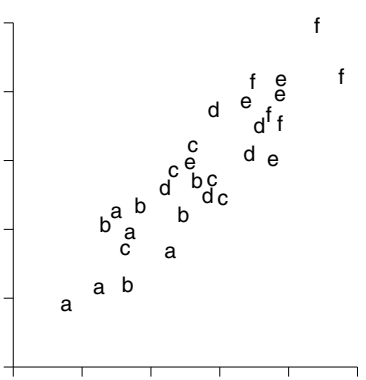
To interpret a correlation coefficient graphically, draw the scatter diagram in your mind's eye so the vertical SD covers the same distance on the page as the vertical SDs in figure 6 on p. 127; and likewise for the horizontal SD. If r for your scatter diagram is 0.40, it will probably show about the same amount of clustering around the diagonal as the one with an r of 0.40 in the figure at the top right. If r is 0.90, it will look like the diagram in the figure at the bottom left. In general, your scatter diagram will match the one that has a similar value for r.

Figure 3. The effect of changing SDs. The two scatter diagrams have the same correlation coefficient of 0.70. The top diagram looks more tightly clustered around the SD line because its SDs are smaller.



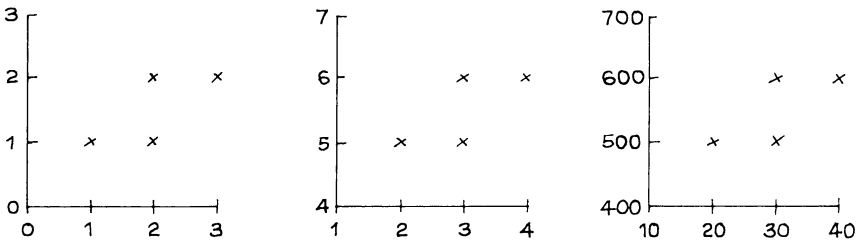
Exercise Set B

1. In the figure below, 6 scatter diagrams are plotted on the same pair of axes; in the first, the points are marked "a"; in the second, "b"; and so forth. For each of the 6 diagrams taken on its own, the correlation is around 0.6. Now take all the points together. For the combined diagram, is the correlation around 0.0, 0.6, or 0.9?

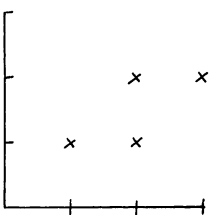


2. The National Health and Nutrition Examination Survey (p. 58) also covers children. In HANES2, at each age from 6 to 11, the correlation between height and weight was just about 0.67. For all the children together, would the correlation between height and weight be just about 0.67, somewhat more than 0.67, or somewhat less than 0.67? Choose one option and explain.

3. Below are three scatter diagrams. Do they have the same correlation? Try to answer without calculating.



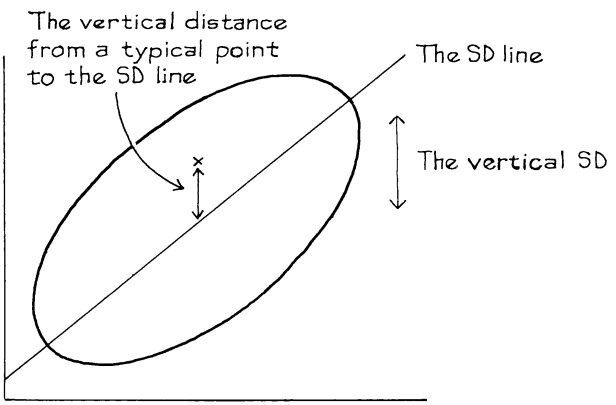
4. Someone hands you the scatter diagram shown below, but has forgotten to label the axes. Can you still calculate r? If so, what is it? Or do you need the labels?



The answers to these exercises are on p. A58.

Technical notes. (i) If r is close to 1, then a typical point is only a small fraction of a vertical SD above or below the SD line. If r is close to 0, then a typical point is above or below the line by an amount roughly comparable in size to the vertical SD: see figure 4. (The "vertical SD" is the SD of the variable plotted on the y-axis.)

Figure 4. The correlation coefficient. As r gets close to 1, the distance of a typical point above or below the SD line becomes a small fraction of the vertical SD.



(ii) The connection between the correlation coefficient and the typical distance above or below the SD line can be expressed mathematically, as follows. The r.m.s. vertical distance to the SD line equals

$$\sqrt{2(1-|r|)} \times \text{ the vertical SD}$$

Take, for example, a correlation of 0.95. Then

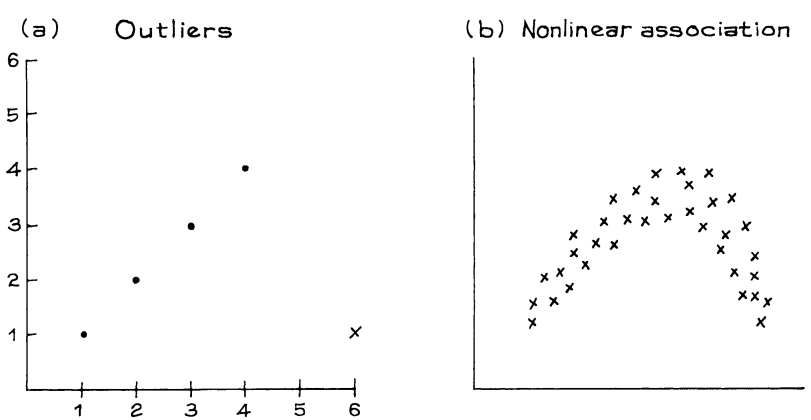
$$\sqrt{2(1-|r|)} = \sqrt{0.1} \approx 0.3$$

So the spread around the SD line is about 30% of a vertical SD. That is why a scatter diagram with $r = 0.95$ shows a fair amount of spread around the line (figure 6 on p. 127). There are similar formulas for the horizontal direction.

3. SOME EXCEPTIONAL CASES

The correlation coefficient is useful for football-shaped scatter diagrams. For other diagrams, r can be misleading. Outliers and non-linearity are problem cases. In figure 5a, the dots show a perfect correlation of 1. The outlier, marked by a cross, brings the correlation down almost to 0. Figure 5a should not be summarized using r. Some people get carried away in pursuit of outliers. However, in any scatter diagram there will be some points more or less detached from the main part of the cloud. These points should be rejected only if there is good reason to do so.

Figure 5. The correlation coefficient can be misleading in the presence of outliers or non-linear association.

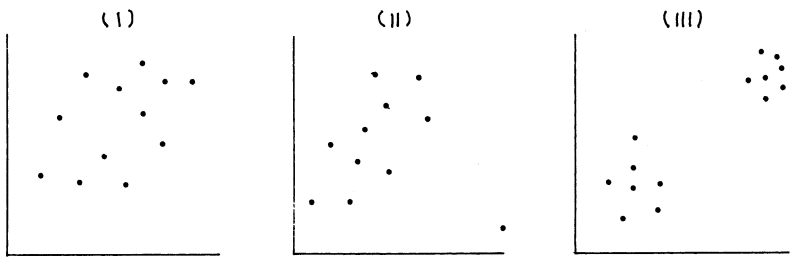


In figure 5b, the correlation coefficient is close to 0, even though the points show a strong association. The reason is that the graph does not look at all like a straight line: as x increases, y rises then falls. This pattern is shown by the association between weight and age for adult men (figure 3 on p. 59). Again, such data should not be summarized using r—the pattern gets lost.

r measures linear association, not association in general.

Exercise Set C

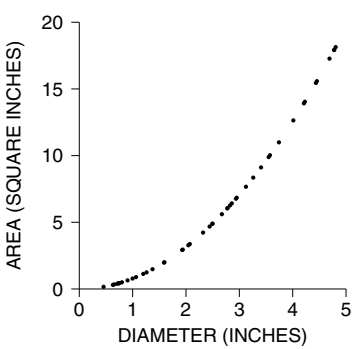
1. Which of the following three scatter diagrams should be summarized by r?



2. A class of 15 students happens to include 5 basketball players. True or false, and explain: the relationship between heights and weights for this class should be summarized using r.

3. A circle of diameter d has area $\frac{1}{4}\pi d^2$. An investigator plots a scatter diagram of area against diameter for a sample of circles with different diameters. (The diagram is shown below.) The correlation coefficient is _____. Fill in the blank, and explain. Options:

-1 nearly -1 nearly 0 nearly 1 1



4. For a certain data set, $r = 0.57$. Say whether each of the following statements is true or false, and explain briefly; if you need more information, say what you need, and why.

   (a) There are no outliers.
   (b) There is a non-linear association.

The answers to these exercises are on p. A58.

4. ECOLOGICAL CORRELATIONS

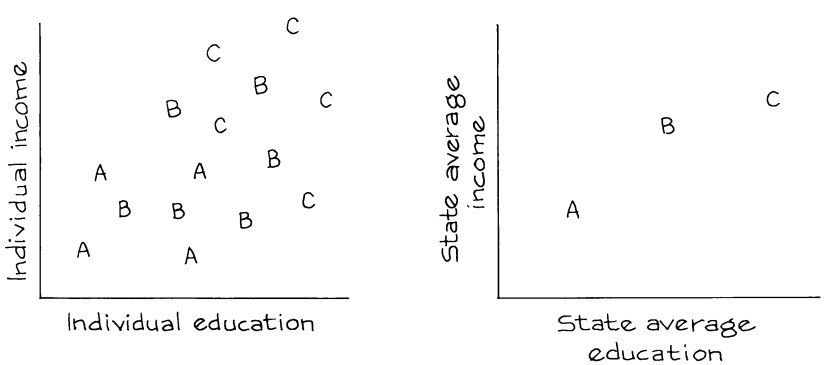
In 1955, Sir Richard Doll published a landmark article on the relationship between cigarette smoking and lung cancer.[2] One piece of evidence was a scatter diagram showing the relationship between the rate of cigarette smoking (per capita) and the rate of deaths from lung cancer in eleven countries. The correla-

tion between these eleven pairs of rates was 0.7, and this was taken as showing the strength of the relationship between smoking and cancer. However, it is not countries which smoke and get cancer, but people. To measure the strength of the relationship for people, it is necessary to have data relating smoking and cancer for individuals rather than countries. Such studies are available, and show that smoking does indeed cause cancer.

The statistical point: correlations based on rates or averages can be misleading. Here is another example. From Current Population Survey data for 2005, you can compute the correlation between income and education for men age 25–64 in the United States: $r \approx 0.42$. For each state (and D.C.), you can compute average educational level and average income. Finally, you can compute the correlation between the 51 pairs of averages: $r \approx 0.70$. If you used the correlation for the states to estimate the correlation for the individuals, you would be way off. The reason is that within each state, there is a lot of spread around the averages. Replacing the states by their averages eliminates the spread, and gives a misleading impression of tight clustering. Figure 6 shows the effect for three states.[3]

> *Ecological* correlations are based on rates or averages. They are often used in political science and sociology. And they tend to overstate the strength of an association. So watch out.

Figure 6. Ecological correlations (based on rates or averages) are usually too big. The panel on the left represents income and education for individuals in three states, labeled A, B, C. Each individual is marked by a letter showing state of residence. The correlation is moderate. The panel on the right shows the averages for each state. The correlation between the averages is almost 1.



Exercise Set D

1. The table at the top of the next page is adapted from Doll and shows per capita consumption of cigarettes in various countries in 1930, and the death rates from lung cancer for men in 1950. (In 1930, hardly any women smoked; and a long period of time is needed for the effects of smoking to show up.)

Country	*Cigarette consumption*	*Deaths per million*
Australia	480	180
Canada	500	150
Denmark	380	170
Finland	1,100	350
Great Britain	1,100	460
Iceland	230	60
Netherlands	490	240
Norway	250	90
Sweden	300	110
Switzerland	510	250
U.S.	1,300	200

(a) Plot a scatter diagram for these data.
(b) True or false: the higher cigarette consumption was in 1930 in one of these countries, on the whole the higher the death rate from lung cancer in 1950. Or can this be determined from the data?
(c) True or false: death rates from lung cancer tend to be higher among those persons who smoke more. Or can this be determined from the data?

2. A sociologist is studying the relationship between suicide and literacy in nineteenth-century Italy.[4] He has data for each province, showing the percentage of literates and the suicide rate in that province. The correlation is 0.6. Does this give a fair estimate of the strength of the association between literacy and suicide?

The answers to these exercises are on p. A59.

5. ASSOCIATION IS NOT CAUSATION

For school children, shoe size is strongly correlated with reading skills. However, learning new words does not make the feet get bigger. Instead, there is a third factor involved—age. As children get older, they learn to read better and they outgrow their shoes. (According to the statistical jargon of chapter 2, age is a confounder.) In the example, the confounder was easy to spot. Often, this is not so easy. And the arithmetic of the correlation coefficient does not protect you against third factors.[5]

> Correlation measures association. But association is not the same as causation.

Example 1. Education and unemployment. During the Great Depression of 1929–1933, better-educated people tended to have shorter spells of unemployment. Does education protect you against unemployment?

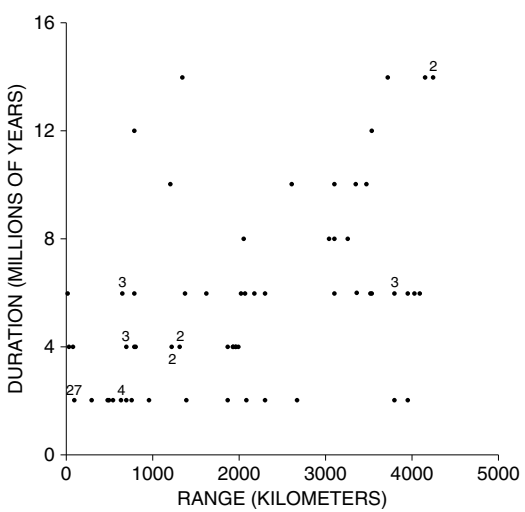
Discussion. Perhaps, but the data were observational. As it turned out, age was a confounding variable. The younger people were better educated, because

the educational level had been going up over time. (It still is.) Given a choice in hiring, employers seemed to prefer younger job-seekers. Controlling for age made the effect of education on unemployment much weaker.[6]

Example 2. Range and duration of species. Does natural selection operate at the level of species? This is a question of some interest for paleontologists. David Jablonski argues that geographical range is a heritable characteristic of species: a species with a wide range survives longer, because if a disaster strikes in one place, the species stays alive at other places.

One piece of evidence is a scatter diagram (figure 7). Ninety-nine species of gastropods (slugs, snails, etc.) are represented in the diagram. The duration of the species—its lifetime, in millions of years—is plotted on the vertical axis; its range is on the horizontal, in kilometers. Both variables are determined from the fossil record. There is a good positive association: r is about 0.64. (The cloud looks formless, but that is because of a few straggling points at the bottom right and the top left.) Does a wide geographical range promote survival of the species?

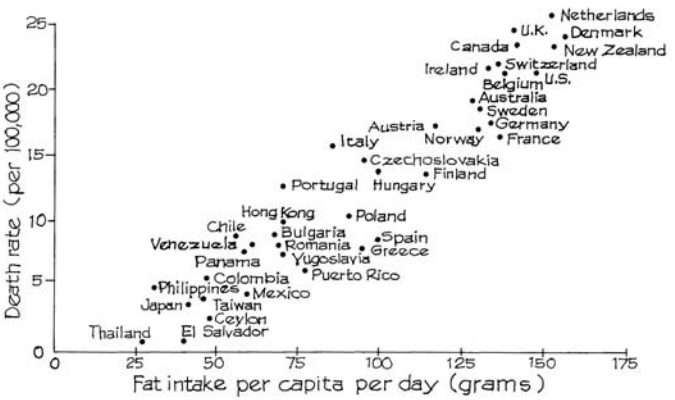
Figure 7. Duration of species in millions of years plotted against geographical range in kilometers, for 99 species of gastropods. Several species can be plotted at the same point; the number of such species is indicated next to the point.



Discussion. A wide range may cause a long lifetime. Or, a long lifetime may cause a wide range. Or, there may be something else going on. Jablonski had his eye on the first possibility. The second one is unlikely, because other evidence suggests that species achieve their ranges very soon after they emerge. But what about the third explanation? Michael Russell and David Lindberg point out that species with a wide geographical range have more chances to be preserved in the fossil record, which can create the appearance of a long lifetime. If so, figure 7 is a statistical artifact.[7] Association is not causation.

Example 3. Fat in the diet and cancer. In countries where people eat lots of fat—like the U.S.—rates of breast cancer and colon cancer are high. See figure 8 for data on breast cancer. This correlation is often used to argue that fat in the diet causes cancer. How good is the evidence?

Figure 8. Death rates from breast cancer plotted against fat in the diet, for a sample of countries.



Note: Age standardized.
Source: K. Carroll, "Experimental evidence of dietary factors and hormone-dependent cancers," *Cancer Research* vol. 35 (1975) p. 3379. Copyright by *Cancer Research*. Reproduced by permission.

Discussion. If fat in the diet causes cancer, then the points in the diagram should slope up, other things being equal. So the diagram is some evidence for the theory. But the evidence is quite weak, because other things aren't equal. For example, the countries with lots of fat in the diet also have lots of sugar. A plot of breast cancer rates against sugar consumption would look just like figure 8, and nobody thinks that sugar causes breast cancer. As it turns out, fat and sugar are relatively expensive. In rich countries, people can afford to eat fat and sugar rather than starchier grain products. Some aspects of the diet in these countries, or other factors in the life-style, probably do cause certain kinds of cancer—and protect against other kinds. So far, epidemiologists can identify only a few of these factors with any real confidence.[8]

Exercise Set E

1. The scatter diagram in figure 7 shows stripes. Why?
2. Is the correlation in figure 8 ecological? How is that relevant to the argument?
3. The correlation between height and weight among men age 18–74 in the U.S. is about 0.40. Say whether each conclusion below follows from the data; explain your answer.
   (a) Taller men tend to be heavier.
   (b) The correlation between weight and height for men age 18–74 is about 0.40.
   (c) Heavier men tend to be taller.

(d) If someone eats more and puts on 10 pounds, he is likely to get somewhat taller.

4. Studies find a negative correlation between hours spent watching television and scores on reading tests.[9] Does watching television make people less able to read? Discuss briefly.

5. Many studies have found an association between cigarette smoking and heart disease. One study found an association between coffee drinking and heart disease.[10] Should you conclude that coffee drinking causes heart disease? Or can you explain the association between coffee drinking and heart disease in some other way?

6. Many economists believe that there is trade-off between unemployment and inflation: low rates of unemployment will cause high rates of inflation, while higher rates of unemployment will reduce the rate of inflation. The relationship between the two variables is shown below for the U.S. in the decade 1960–69. There is one point for each year, with the rate of unemployment that year shown on the x-axis, and the rate of inflation shown on the y-axis. The points fall very close to a smooth curve known as the *Phillips Curve*. Is this an observational study or a controlled experiment? If you plotted the points for the 1970s or the 1950s, would you expect them to fall along the curve?

The Phillips curve for the 1960s:
Economic Report of the President (1975)

The answers to these exercises are on p. A59.

6. REVIEW EXERCISES

Review exercises may cover material from previous chapters.

1. When studying one variable, you can use a graph called a _____. When studying the relationship between two variables, you can use a graph called a _____.

2. True or false, and explain briefly:
   (a) If the correlation coefficient is -0.80, below-average values of the dependent variable are associated with below-average values of the independent variable.
   (b) If y is usually less than x, the correlation coefficient between x and y will be negative.

3. In each case, say which correlation is higher, and explain briefly. (Data are from a longitudinal study of growth.)
   (a) Height at age 4 and height at age 18, height at age 16 and height at age 18.
   (b) Height at age 4 and height at age 18, weight at age 4 and weight at age 18.
   (c) Height and weight at age 4, height and weight at age 18.

4. An investigator collected data on heights and weights of college students; results can be summarized as follows.

	Average	*SD*
Men's height	70 inches	3 inches
Men's weight	144 pounds	21 pounds
Women's height	64 inches	3 inches
Women's weight	120 pounds	21 pounds

The correlation coefficient between height and weight for the men was about 0.60; for the women, it was about the same. If you take the men and women together, the correlation between height and weight would be _____ .

just about 0.60 somewhat lower somewhat higher

Choose one option, and explain briefly.

5. A number is missing in each of the data sets below. If possible, fill in the blank to make r equal to 1. If this is not possible, say why not.

(a)

x	y
1	1
2	3
2	3
4	–

(b)

x	y
1	1
2	3
3	4
4	–

6. A computer program prints out r for the two data sets shown below. Is the program working correctly? Answer yes or no, and explain briefly.

(i)

x	y
1	2
2	1
3	4
4	3
5	7
6	5
7	6

$r = 0.8214$

(ii)

x	y
1	5
2	4
3	7
4	6
5	10
6	8
7	9

$r = 0.7619$

7. In 1910, Hiram Johnson entered the California gubernatorial primaries. For each county, data are available to show the percentage of native-born Americans in that county, as well as the percentage of the vote for Johnson. A

political scientist calculated the correlation between these percentages.[11] It is 0.5. Is this a fair measure of the extent to which "Johnson received native, as opposed to immigrant, support?" Answer yes or no, and explain briefly.

8. For women age 25 and over in the U.S. in 2005, the relationship between age and educational level (years of schooling completed) can be summarized as follows:[12]

$$\text{average age} \approx 50 \text{ years}, \quad \text{SD} \approx 16 \text{ years}$$
$$\text{average ed. level} \approx 13.2 \text{ years}, \quad \text{SD} \approx 3.0 \text{ years}, \quad r \approx -0.20$$

True or false, and explain: as you get older, you become less educated. If this statement is false, what accounts for the negative correlation?

9. At the University of California, Berkeley, Statistics 2 is a large lecture course with small discussion sections led by teaching assistants. As part of a study, at the second-to-last lecture one term, the students were asked to fill out anonymous questionnaires rating the effectiveness of their teaching assistants (by name), and the course, on the scale

1	2	3	4	5
poor	fair	good	very good	excellent

The following statistics were computed.

- The average rating of the assistant by the students in each section.
- The average rating of the course by the students in each section.
- The average score on the final for the students in each section.

Results are shown below (sections are identified by letter). Draw a scatter diagram for each pair of variables—there are three pairs—and find the correlations.

Section	*Ave. rating of assistant*	*Ave. rating of course*	*Ave. score on final*
A	3.3	3.5	70
B	2.9	3.2	64
C	4.1	3.1	47
D	3.3	3.3	63
E	2.7	2.8	69
F	3.4	3.5	69
G	2.8	3.6	69
H	2.1	2.8	63
I	3.7	2.8	53
J	3.2	3.3	65
K	2.4	3.3	64

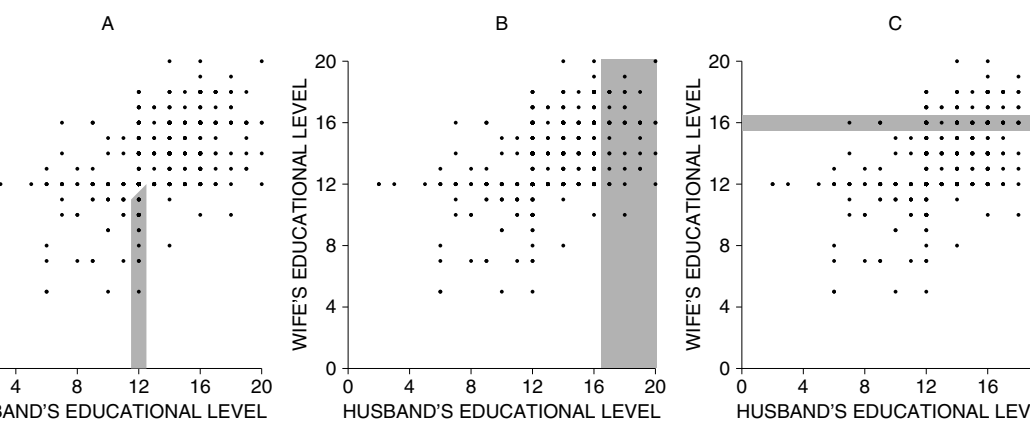
The data are section averages. Since the questionnaires were anonymous, it was not possible to link up student ratings with scores on an individual basis. Student ability may be a confounding factor. However, controlling for pretest results turned out to make no difference in the analysis.[13] Each assistant taught one section. True or false, and explain:

(a) On the average, those sections that liked their TA more did better on the final.
(b) There was almost no relationship between the section's average rating of the assistant and the section's average rating of the course.
(c) There was almost no relationship between the section's average rating of the course and the section's average score on the final.

10. In a study of 2005 Math SAT scores, the Educational Testing Service computed the average score for each of the 51 states, and the percentage of the high-school seniors in that state who took the test.[14] (For these purposes, D.C. counts as a state.) The correlation between these two variables was equal to -0.84.
    (a) True or false: test scores tend to be lower in the states where a higher percentage of the students take the test. If true, how do you explain this? If false, what accounts for the negative correlation?
    (b) In Connecticut, the average score was only 517. But in Iowa, the average was 608. True or false, and explain: the data show that on average, the schools in Iowa are doing a better job at teaching math than the schools in Connecticut.

11. As part of the study described in exercise 10, the Educational Testing Service computed the average Verbal SAT score for each state, as well as the average Math SAT score for each state. (Again, D.C. counts as a state.) The correlation between these 51 pairs of averages was 0.97. Would the correlation between the Math SAT and the Verbal SAT—computed from the data on all the individuals who took the tests—be larger than 0.97, about 0.97, or less than 0.97? Explain briefly.

12. Shown below is a scatter diagram for educational levels (years of schooling completed) of husbands and wives in South Carolina, from the March 2005 Current Population Survey.
    (a) The points make vertical and horizontal stripes. Why?



(b) There were 530 couples in the sample, and there is a dot for each couple. But if you count, there are only 104 dots in the scatter diagram. How can that be? Explain briefly.

(c) Three areas are shaded. Match the area with the description. (One description will be left over.)

  (i) Wife completed 16 years of schooling.
  (ii) Wife completed more years of schooling than husband.
  (iii) Husband completed more than 16 years of schooling.
  (iv) Husband completed 12 years of schooling and wife completed fewer years of schooling than husband.

7. SUMMARY

1. The correlation coefficient is a pure number, without units. It is not affected by

- interchanging the two variables,
- adding the same number to all the values of one variable,
- multiplying all the values of one variable by the same positive number.

2. The correlation coefficient measures clustering around a line, relative to the SDs.

3. The correlation coefficient can be misleading in the presence of outliers or non-linear association. Whenever possible, look at the scatter diagram to check for these problems.

4. *Ecological* correlations, which are based on rates or averages, tend to overstate the strength of associations for individuals.

5. Correlation measures association. But association does not necessarily show causation. It may only show that both variables are simultaneously influenced by some third variable.